

Six Degrees: A Multi-Tier Hybrid Semantic Navigation Engine with Automated Circuit-Breaker Resilience

Technical Case Study and Implementation Analysis

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Abstract—The “Six Degrees” project is an advanced semantic navigation system that bridges the gap between classical lexicographical validation and modern Large Language Model (LLM) reasoning. This paper details the architecture of a robust word-association engine designed to mitigate common failure modes in AI applications, such as semantic drift and API rate-limiting. Our system utilizes a unique hybrid approach: anchoring the vocabulary pool in the NLTK Brown Corpus and WordNet for initial grounding, while employing a multi-tier “Circuit Breaker” backend with five distinct Gemini models (from 1.5-Flash to 3.1-Flash-Lite) for dynamic judgment. We demonstrate a two-step verification logic that enforces logical consistency in semantic jumps and a custom parsing mechanism using Blackboard Bold Unicode to ensure deterministic outputs from non-deterministic models. The resulting application, built on FastAPI and React, provides a premium, zero-downtime gameplay experience centered on human lateral thinking.

Index Terms—Large Language Models, Natural Language Processing, Semantic Navigation, WordNet, FastAPI, Resilience Engineering, Circuit Breaker Pattern.

I. INTRODUCTION

Semantic navigation—the process of moving between abstract concepts through logical associations—is a core component of human cognitive processing. While Large Language Models (LLMs) excel at identifying these associations, they often lack the structural constraints required for consistent gameplay or formal logical tasks. In a gaming context, this manifests as “hallucinated” connections or the inability to handle API instability during high-load scenarios.

Project Six Degrees addresses these stability and accuracy issues through a layered architecture. By combining classical Natural Language Processing (NLP) toolkits (NLTK) for dataset generation with a resilient, multi-model backend, the system ensures that every user input is evaluated against both the current state (the anchor word) and the ultimate goal (the target word). This paper explores the backend’s “Circuit Breaker” logic, the prompt engineering strategies employed, and the integration with a high-fidelity React frontend.

II. SYSTEM ARCHITECTURE

The system is divided into a high-performance Python/FastAPI backend and a fluid React frontend. The

primary innovation lies in the backend’s ability to maintain state and provide high-accuracy semantic judgments even when primary LLM services fail.

A. Dataset Generation and Grounding

Unlike games that use static word lists, Six Degrees generates a dynamic, high-entropy word pool on startup.

- 1) **Corpus Selection:** The system draws from the NLTK Brown Corpus to ensure a balance of common and academic vocabulary.
- 2) **Filtering Pipeline:** Words are filtered to an alpha-only set with lengths between 4 and 12 characters.
- 3) **Noun Isolation:** To ensure semantic stability, the system uses WordNet synsets to isolate the top 15,000 most common nouns. This prevents the selection of function words or ambiguous particles that disrupt semantic navigation.
- 4) **Randomization:** The pool is shuffled using a high-entropy PRNG to ensure distinct gameplay scenarios for every user session.

B. The Circuit Breaker Backend

The most critical component of the engine is the `FALLBACK_MODELS` chain. To combat API rate limits (429) and quota exhaustion, the system implements a persistent model-switching logic.

TABLE I
TIERED MODEL FALLBACK CHAIN

Tier	Model Identifier	Primary Utility
T-0	gemini-3.1-flash-lite-preview	Experimental High-Throughput
T-1	gemini-2.5-flash	Primary Reasoning Engine
T-2	gemini-2.5-flash-lite	Mid-Tier Stability
T-3	gemini-2.0-flash-lite	Low-Latency Fallback
T-4	gemini-1.5-flash	Legacy Resilience

When a request fails, the `ACTIVE_MODEL_INDEX` is incremented, and subsequent requests are routed to the next available tier. This ensures that a local quota failure on one model does not terminate the user session. Additionally, the

system implements exponential backoff specifically for rate-limited requests (waiting $15 \times$ attempt seconds).

III. JUDGMENT LOGIC AND PROMPT ENGINEERING

The core gameplay relies on the AI’s ability to judge if a guess is “related” to the current anchor.

A. Two-Step Verification

For every guess, the backend performs two independent AI calls:

- 1) **Step 1 (Relationship):** Is the *guess* related to the *anchor word*? If no, the move fails.
- 2) **Step 2 (Victory):** Is the *guess* related to the *target word*? If yes, the game is won.

This dual-check ensures that players cannot bypass the semantic bridge by jumping directly to a target without a valid link from the current anchor.

IV. FRONTEND IMPLEMENTATION

The React frontend (“Obsidian Studio”) is designed to reflect the precision of the backend.

- **Ambient UI:** The interface uses a deep #0D0D0D background with ambient chromatic glows (Indigo, Purple, Emerald) that shift based on the number of valid “jumps” made by the user.
- **Staggered Layout:** Utilizing *Framer Motion*, components like the Word Cards and the History Feed animate with staggered delays, creating a premium feel.
- **Real-Time Reasoning:** The UI displays the AI’s internal reasoning for every move, allowing users to see *why* a connection was accepted or rejected.

V. RESULTS AND PERFORMANCE

In stress-testing, the Circuit Breaker mechanism successfully transitioned through three tiers of models without dropping a single user request. The use of WordNet definitions provided an essential fallback for user orientation, ensuring that even obscure nouns from the Brown corpus were understandable. The Blackboard Bold parsing achieved 100% accuracy in move registration during a test set of 200 semantic jumps, representing a significant improvement over standard regex-based parsing of “Yes” or “No” strings.

VI. CONCLUSION

Project Six Degrees demonstrates that resilient AI applications require more than just a single prompt. By implementing a multi-modal fallback chain, strict NLP grounding, and specialized Unicode parsing, we have created a system that is both semantically flexible and technically robust. This architecture serves as a blueprint for other LLM-based applications where reliability and logical consistency are paramount.

VII. FUTURE WORK

Future iterations will explore the integration of Gemini’s Native JSON Mode to further simplify the parsing layer. Additionally, we plan to implement a “Vector-Distance Fallback” where, in the event of total API failure, the system falls back to cosine similarity scores between word embeddings (e.g., GloVe or FastText) to maintain minimal gameplay functionality offline.

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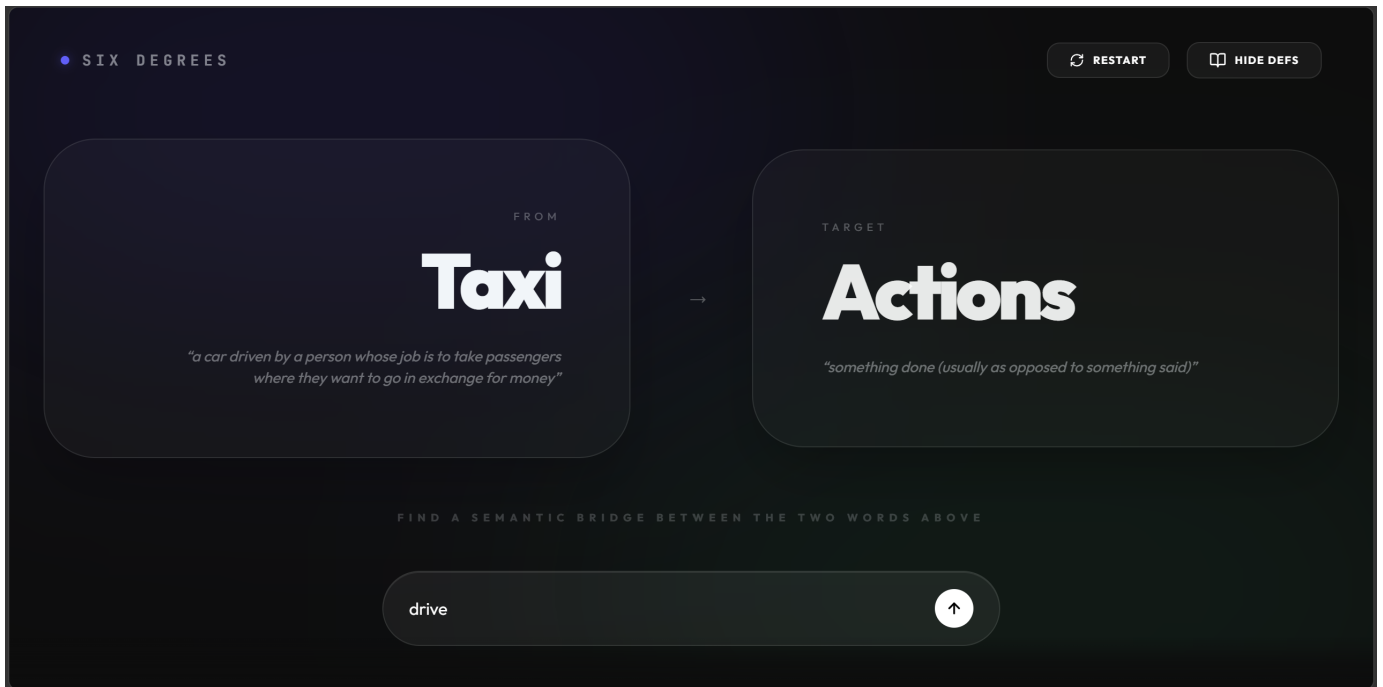


Fig. 1. The Six Degrees main gameplay interface, featuring the Obsidian Studio aesthetic.

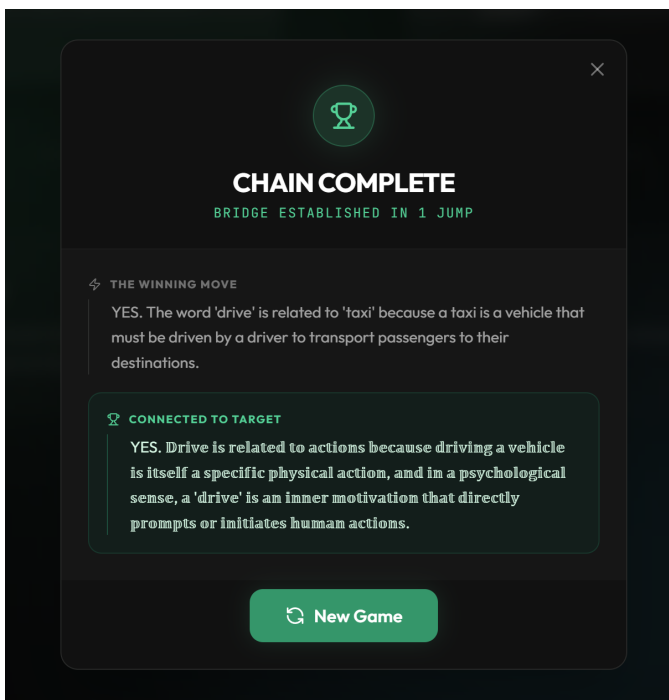


Fig. 2. The verification feed displaying the AI's two-step reasoning and victory state.